# INTRODUCTION TO VPNs

## Introduction

A Virtual Private Network (VPN) creates a logical private connection over an untrusted or shared network such as the Internet. VPNs are commonly used to connect company branches and to let employees securely access an organisation's internal network from remote locations.

## Types of VPNs

### 1. Site-to-site VPN

A site-to-site VPN connects two or more complete networks, such as the Ahmedabad and Mumbai branches of a company. VPN gateways (usually routers or firewalls) establish the connection, so users at either site can access permitted resources at the other site without starting a VPN session on every individual PC. IPsec is commonly used to encrypt site-to-site traffic.

### 2. Remote access VPN

A remote access VPN connects an individual user's device to an organisation's network. The user runs VPN client software and authenticates with credentials, certificates or multi-factor authentication. It is commonly used by employees working from home or while travelling.

## Task 1: Create Two Branch Networks and Configure IP Addressing

Example addressing plan (use the same addresses in your Packet Tracer topology):

| Device / Interface | IP address | Subnet mask | Default gateway |
|---|---|---|---|
| PC-A | 192.168.10.10 | 255.255.255.0 | 192.168.10.1 |
| R1 G0/0 (LAN) | 192.168.10.1 | 255.255.255.0 | — |
| R1 G0/1 (WAN) | 198.51.100.1 | 255.255.255.252 | — |
| ISP G0/0 | 198.51.100.2 | 255.255.255.252 | — |
| ISP G0/1 | 203.0.113.1 | 255.255.255.252 | — |
| R2 G0/1 (WAN) | 203.0.113.2 | 255.255.255.252 | — |
| R2 G0/0 (LAN) | 192.168.20.1 | 255.255.255.0 | — |
| PC-B | 192.168.20.10 | 255.255.255.0 | 192.168.20.1 |

Connect R1 and R2 to an ISP router as shown in the diagram. Interface names may differ by router model; use the actual GigabitEthernet/FastEthernet ports in your file.

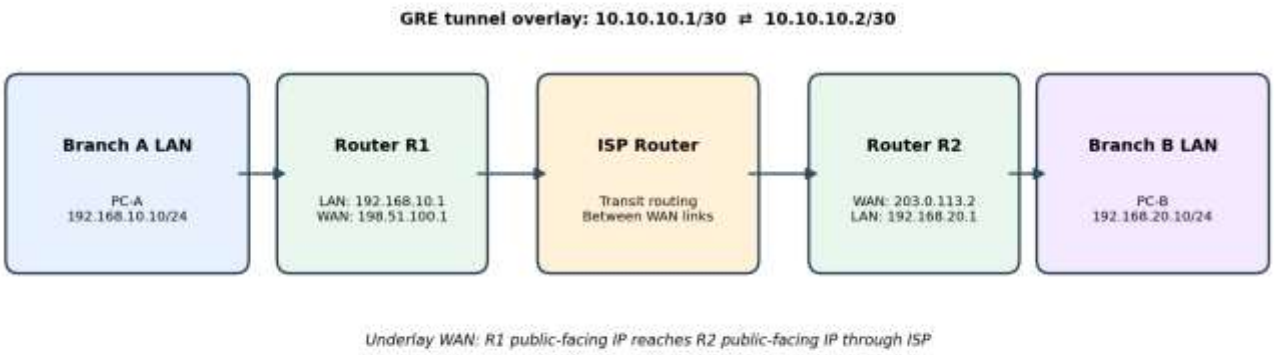



*Figure 1: Two branch LANs connected through an ISP WAN, with a GRE tunnel overlay.*

## Configure the ISP Router

On the ISP router:

```
enable
configure terminal
interface gigabitEthernet 0/0
 ip address 198.51.100.2 255.255.255.252
 no shutdown
exit
interface gigabitEthernet 0/1
 ip address 203.0.113.1 255.255.255.252
 no shutdown
end
show ip interface brief
```

## Configure Router R1 (Branch A)

```
enable
configure terminal
interface gigabitEthernet 0/0
 ip address 192.168.10.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet 0/1
 ip address 198.51.100.1 255.255.255.252
 no shutdown
exit
ip route 203.0.113.0 255.255.255.252 198.51.100.2
```

```
end
show ip interface brief
ping 198.51.100.2
```

## Configure Router R2 (Branch B)

```
enable
configure terminal
interface gigabitEthernet 0/0
 ip address 192.168.20.1 255.255.255.0
 no shutdown
exit
interface gigabitEthernet 0/1
 ip address 203.0.113.2 255.255.255.252
 no shutdown
exit
ip route 198.51.100.0 255.255.255.252 203.0.113.1
end
show ip interface brief
ping 203.0.113.1
```

## Configure the PCs

On PC-A, open Desktop > IP Configuration and enter IP 192.168.10.10, mask 255.255.255.0, gateway 192.168.10.1. On PC-B, enter IP 192.168.20.10, mask 255.255.255.0, gateway 192.168.20.1. Before creating the tunnel, each PC should be able to ping its own branch router.

## Task 2: Configure the Site-to-site GRE Tunnel

GRE (Generic Routing Encapsulation) creates a logical tunnel between the routers. In this lab, the tunnel uses the public-facing WAN interface addresses as its source and destination. The tunnel IP addresses form a separate point-to-point network.

### R1 tunnel configuration

```
enable
configure terminal
interface tunnel 0
 ip address 10.10.10.1 255.255.255.252
 tunnel source gigabitEthernet 0/1
 tunnel destination 203.0.113.2
 no shutdown
exit
ip route 192.168.20.0 255.255.255.0 10.10.10.2
end
```

### R2 tunnel configuration

```
enable
```

```
configure terminal
interface tunnel 0
 ip address 10.10.10.2 255.255.255.252
 tunnel source gigabitEthernet 0/1
 tunnel destination 198.51.100.1
 no shutdown
exit
ip route 192.168.10.0 255.255.255.0 10.10.10.1
end
```

Important: the WAN underlay must already be able to route between 198.51.100.1 and 203.0.113.2. If your topology uses another ISP design, adjust the WAN IPs, next hops, tunnel source and tunnel destination accordingly.

### Verify the tunnel and test end-to-end connectivity

```
show interfaces tunnel 0
show ip interface brief
show ip route
show running-config
ping 10.10.10.2
ping 192.168.20.10
```

From PC-A, open Desktop > Command Prompt and run `ping 192.168.20.10`. From PC-B, run `ping 192.168.10.10`. Successful replies confirm routing between the branch LANs through the GRE tunnel.

## Task 3: Remote Access VPN Simulation in Packet Tracer

Remote-access VPN support depends on the Packet Tracer version and device model. A common Packet Tracer classroom simulation uses an ASA 5505 security appliance as the VPN gateway and a PC with a built-in VPN client. This is a separate demonstration from the router-to-router GRE tunnel above.

General simulation procedure (where the VPN Client feature is available):

1. Add an ASA 5505 between the outside/WAN network and the branch LAN. Configure its inside and outside interfaces with valid IP addresses and make sure the PC can reach the ASA's outside address.
2. On the ASA, use the Packet Tracer-supported VPN configuration method or VPN wizard to enable remote-access IPsec VPN. Configure a VPN group name, pre-shared key, user account and password, and an address pool for remote clients.
3. On the remote PC, open Desktop > VPN (or the VPN Client tab, depending on the Packet Tracer release).
4. Enter the ASA outside/server IP address, the configured group name and pre-shared key if requested, and the remote-access username and password.
5. Select Connect. After the client reports a successful connection, test access to an allowed inside-LAN host.

If your Packet Tracer PC does not show a VPN tab, or your ASA/router image does not offer remote-access VPN configuration, that feature is not available in that version or device model. Do not claim that a remote-access VPN was successfully configured; record the limitation and ask the instructor which supported device/version to use. The GRE tunnel in Task 2 demonstrates tunnelling but does not encrypt traffic.

## Task 4: Site-to-site VPN vs. Remote Access VPN

| Point | Site-to-site VPN | Remote access VPN |
|---|---|---|
| Connection | Connects entire networks/branch LANs. | Connects one user device to an organisation network. |
| Typical endpoint | VPN gateways such as routers or firewalls. | VPN client on a laptop/mobile and a VPN |

| | | |
|---|---|---|
| | | gateway. |
| User experience | Usually always-on or gateway-managed; users do not connect individually. | User normally signs in and starts a VPN session. |
| Advantages | 1. Many branch users share one gateway connection. 2. Useful for permanent branch-to-branch connectivity. 3. Centralised gateway management. | 1. Supports work from home and travel. 2. Individual user authentication and access policies. 3. Flexible for changing locations. |
| Disadvantages | 1. Gateway and WAN setup can be complex. 2. Requires compatible gateway configuration at each site. 3. A gateway/link failure can affect many users. | 1. Each user needs a client and credentials. 2. User experience depends on Internet quality. 3. Help-desk and account management increase with many users. |

## Task 5: Step-by-step VPN Configuration Summary

6. Create two LANs: Branch A 192.168.10.0/24 and Branch B 192.168.20.0/24.
7. Configure router LAN and WAN interfaces and use `no shutdown` on each active interface.
8. Configure the ISP router so it routes between the two WAN subnets.
9. Confirm WAN reachability between the router public-facing interface addresses.
10. On both routers, create `interface tunnel 0`, assign tunnel IP addresses, and set matching tunnel source and destination endpoints.
11. Add static routes for each remote LAN through the opposite tunnel IP address.
12. Verify the tunnel interface and routing table, then ping the remote PC from each branch.

## Important Technical Note

The GRE tunnel configured in this assignment provides encapsulation and a logical path between the two routers, but GRE by itself does not provide encryption, authentication or confidentiality. For a production secure site-to-site VPN, use IPsec (often GRE over IPsec when both routing flexibility and encryption are required). A Packet Tracer GRE-only lab should therefore be described as a basic VPN-style tunnel demonstration, not as an encrypted VPN.

## Conclusion

A site-to-site VPN connects complete branch networks, while a remote access VPN connects an individual user to a private network. In this practical, GRE tunnel interfaces and static routes provide connectivity between two branch LANs. A remote-access VPN uses a client and a VPN gateway with user authentication. For real security, VPN traffic should be protected with a suitable encryption protocol such as IPsec.